

AI-driven disinformation: a framework for organizational preparation and response

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Abstract

Purpose – Disinformation, false information designed with the intention to mislead, can significantly damage organizational operation and reputation, interfering with communication and relationship management in a wide breadth of risk and crisis contexts. Modern digital platforms and emerging technologies, including artificial intelligence (AI), introduce novel risks in crisis management (Guthrie and Rich, 2022). Disinformation literature in security and computer science has assessed how previously introduced technologies have affected disinformation, demanding a systematic and coordinated approach for sustainable counter-disinformation efforts. However, there is a lack of theory-driven, evidence-based research and practice in public relations that advises how organizations can effectively and proactively manage risks and crises driven by AI (Guthrie and Rich, 2022).

Design/methodology/approach – As a first step in closing this research-practice gap, the authors first synthesize theoretical and technical literature characterizing the effects of AI on disinformation. Upon this review, the authors propose a conceptual framework for disinformation response in the corporate sector that assesses (1) technologies affecting disinformation attacks and counterattacks and (2) how organizations can proactively prepare and equip communication teams to better protect businesses and stakeholders.

Findings – This research illustrates that future disinformation response efforts will not be able to rely solely on detection strategies, as AI-created content quality becomes more and more convincing (and ultimately, indistinguishable), and that future disinformation management efforts will need to rely on content influence rather than volume (due to emerging capabilities for automated production of disinformation). Built upon these fundamental, literature-driven characteristics, the framework provides organizations actor-level and content-level perspectives for influence and discusses their implications for disinformation management.

Originality/value – This research provides a theoretical basis and practitioner insights by anticipating how AI technologies will impact corporate disinformation attacks and outlining how companies can respond. The proposed framework provides a theory-driven, practical approach for effective, proactive disinformation management systems with the capacity and agility to detect risks and mitigate crises driven by evolving AI technologies. Together, this framework and the discussed strategies offer great value to forward-looking disinformation management efforts. Subsequent research can build upon this framework as AI technologies are deployed in disinformation campaigns, and practitioners can leverage this framework in the development of counter-disinformation efforts.

Keywords Disinformation, Artificial intelligence, Social-mediated crisis communication, Dissemination, Influence

Paper type Research paper

Introduction

As news and popular media are increasingly engaged in the debate of the “post-truth” era (Keyes, 2004), truth and reality, as socially constructed (Berger and Luckmann, 1967), have been unprecedentedly threatened by disinformation, the impact and danger of which is magnified by social media and emerging technologies. Disinformation, false information designed to mislead (e.g., Fallis, 2009; Fallis, 2015; Ryan *et al.*, 2020; Guess and Lyons, 2020), can significantly damage organizational operation and reputation, interfering with organization-public communication and relationship management in a wide breadth of risk and crisis contexts (European Commission, Directorate-General for Communications Networks, Content and Technology, 2018). Disinformation campaigns have posed an ongoing threat and imminent danger to public health, national security, and democracy across the



globe (Fallis, 2015). Countless instances of disinformation have already occurred and are capable of significant impact without the support of sophisticated technologies (e.g., Yankoski *et al.*, 2021); however, emerging technologies, including artificial intelligence (AI), introduce novel risks in crisis communication and management (Guthrie and Rich, 2022), and there exists great potential for integrating AI in disinformation operations.

Practitioners do not yet have a full understanding of the interplay between AI technologies and disinformation campaigns; factors such as a platform's dissemination algorithms and the flow of content across platforms create new organizational vulnerabilities, as traditional strategies for monitoring and mitigation risk missing essential information for optimized management efforts (Hurley, 2023). Practitioners are noting the impact of recent AI technologies, such as GPT-4 (OpenAI, 2023), in mediating online environments and transforming consumer search behaviors (PRNEWS, 2023). Such AI technologies have important implications for practitioner workflows and processes related to corporate communication and disinformation management, and analysis of AI-driven cases is a critical aspect of disinformation research.

As expressed by PRNEWS (2023), "AI-generated content will increase exponentially" and thus, practitioners must adjust their communication strategies. Literature across disciplines, including security and computer science, has assessed how previously introduced technologies have affected disinformation, such as digital editing of images, the impersonation of websites, and manipulation of Wikipedia entries (Fallis, 2015). Scholars have examined the emergence of deepfakes, micro-targeting, and fake profiles (Kertysova, 2018; Korshunov and Marcel, 2018). The *Truth, Lies, and Automation* report by the Center for Security and Emerging Technology (CSET) (Buchanan *et al.*, 2021) further examined the capabilities of automation-based AI, when deployed by human-machine teams, to amplify disinformation via narrative reiteration, elaboration, manipulation, seeding, wedging, and persuasion. In reference to such technologies, Hurley (2023) notes, "AI technologies such as GPT will be utilized for attacks and must be part of the corporate communications counter-attack toolkit." The current and future stakes are so high that any effective and sustainable counter-disinformation efforts demand a more systematic and coordinated approach.

In early discussion regarding these technological developments and how to navigate AI-mediated environments, practitioners are looking toward the increased importance of "building mindshare", or strong branding and cognitive relevance among target audiences, and direct communication with key publics (PRNEWS, 2023). However, despite the increasing volume of research and insights on disinformation and AI, there is a lack of a theory-driven, evidence-based framework that advises how organizations and businesses might effectively and proactively manage risks and crises caused by AI-driven disinformation attacks from an organizational communication and public relations focus (Guthrie and Rich, 2022). It is critical for organizations to devise more proactive disinformation management systems and more effective strategies in order to protect their assets and safeguard public health and safety. Most recent development in social-mediated crisis communication (SMCC) research has also made urgent arguments for more effective management systems of misinformation and disinformation (Austin *et al.*, 2021; Liu *et al.*, 2021; Lu and Jin, 2020; Lu *et al.*, 2022; van der Meer and Jin, 2022) in the competitive and conflicting digital media ecosystem.

The SMCC model is one of the leading theories used by scholars and practitioners in guiding their crisis management efforts in a social-mediated communication environment (Jin *et al.*, 2022; Liu *et al.*, 2023). For the SMCC model to add further value in organizational disinformation management, especially in light of the disruptive role AI plays in (dis)information generation and spread, the key elements and the shape of (dis)information flow need to be extended beyond the existing SMCC model. Most recently, Jin *et al.* (2022) and Liu *et al.* (2023) posited a set of new directions for enhancing the SMCC model, which goes beyond

the traditional social media platform consideration and into the emerging spaces with AI-generated disinformation as one of the major threats to organizations and their stakeholders. Based on a broad review of the literature on AI and grounded in the SMCC model, this study aims at enriching the SMCC scholarship in AI and disinformation arenas as called for by scholars and practitioners alike (Guthrie and Rich, 2022). To echo the recent call for SMCC extension, one of the main goals and contributions of this study is to develop an AI and disinformation focused framework, reflecting the SMCC core (i.e., how organizations can capture the map of [dis]information flow as created, shared and received by various entities in the digital space) and expanding its capability to describe, explain, and predict social-mediated crisis communication in an increasingly challenging digital environment.

As a first attempt in closing this research-practice gap from the perspective of organizational communication management, grounded in existing AI and disinformation research and the SMCC model, along the prevention, preparation, response, and revisit stages of managing ongoing crises (Coombs, 2019), we propose a conceptual framework that guides (1) the communication technological practices and trends in disinformation attacks and counterattacks and (2) how emerging AI technologies are transforming disinformation (e.g., how disinformation is uniquely propagated) and how organizations can be proactively prepared for it and equip their communication and public relations teams to better protect businesses and stakeholders. While impactful disinformation can continue to be created without sophisticated technologies, we focus specifically on AI-driven cases to investigate their implications on broader media environments.

The foundation of the new framework is laid by first synthesizing theoretical and technical takeaways essential to the organizational communication management context, focusing on identifying and characterizing the effects of AI on disinformation: content creation, dissemination, detection and management strategies, and perception (e.g., Kertysova, 2018). We then map out possible levers, or tools to drive strategies which public relations practitioners might consider utilizing in responding proactively to evolving AI and deep learning technologies that allow for the creation and dissemination of more realistic and widespread disinformation, presenting tremendous challenges to organizations especially during crises in which high-stakes information must be effectively communicated (Vafeiadis *et al.*, 2019) to stakeholders.

This new framework can be applied to (1) understanding disinformation driven or enhanced by AI technologies and (2) guiding the strategic planning and implementation of effective and proactive disinformation counterattacks. Theoretical and practical implications for public relations scholars and communication management practitioners are discussed for building more effective and proactive disinformation management systems as AI-driven disinformation campaigns may continue to evolve along with increasingly complex and uncertain online ecosystems.

Literature review

Disinformation, intentionally inaccurate content developed with the purpose of misleading recipients (e.g., Fallis, 2009; Fallis, 2015; Ryan *et al.*, 2020; Guess and Lyons, 2020), is not a new challenge, as humans have had to exercise discernment in information processing throughout history (Fallis, 2009; Ryan *et al.*, 2020); however, modern technologies introduce new challenges related to the scale of misleading information and algorithmically-backed feedback loops and amplification (Ryan *et al.*, 2020). In this review, we draw upon cross-disciplinary literature to identify key developments in AI technologies and to discuss how these technologies are affecting communication and disinformation processes on social media platforms, crucial channels for modern organizational communication efforts and consumer engagement. We begin by assessing the current technological landscape, identifying

literature that reveals state-of-the-art AI developments; given the various modalities of digital content, we discuss these cutting-edge AI developments as they relate to each digital modality respectively. Because the scope of this research is to investigate organizational strategies to respond to evolving disinformation, we begin with a user-focused perspective discussing psychological processes and vulnerabilities affecting disinformation consumption, and we conclude with an overview of organizational approaches that have been developed in response to disinformation in media environments thus far.

Social media algorithmic landscape

Globally in 2021, approximately 575,000 Tweets were posted per minute, and approximately 167,000,000 TikTok videos were watched per minute ([Statista Research Department, 2022](#)). With the excessive amount of (frequently contradictory) information available online today, the algorithms underlying media platforms play a crucial role as users navigate and receive information. Platform-level algorithmic designs characterize digital environments, specifically recommendation algorithms (e.g., TikTok “For You Page”, Instagram “Explore”, trending content on Twitter) and search engines (to navigate content, e.g., users, posts). The algorithmic nuances of these platforms are not publicly available (with the only public information coming from statements from the platforms, whistleblowers, and academic research), and publics are largely unaware of how algorithms affect content dissemination ([Lewandowsky and Kozyreva, 2022](#); [Reviglio and Agosti, 2020](#)); however, key mechanisms drive content discovery, such as personalization (driven by individual data), trendiness surrounding a topic, social networks, and level of engagement (e.g., [Twitter, n.d.](#)). Thus, dissemination at scale can be accomplished through a combination of the following algorithmic levers: number of accounts, amount of content for a given account, and level of engagement on a piece of content (in the form of views, shares, likes, comments, etc.).

Over the past decade, media platforms have faced increasing social and governmental demand for better disinformation policy, with challenges exacerbated by the COVID-19 infodemic, or barrage of vast amounts of information (both factual and misleading), ([World Health Organization, 2020](#)) and the Russia-Ukraine war; recently, the European Union proposed the Digital Services Act, which would introduce regulation related to increased responsibility for illicit content, targeting, and algorithmic transparency ([Mackrael, 2022](#)). While platforms have expressed their commitment, begun to implement policies, and heavily invested in mitigating the spread of disinformation, disinformation continues to be a pressing challenge and these policies continue to evolve. [Djordjevic \(2020\)](#) reviews the strategies social media companies have implemented in combating fake news. As of May 2022, Twitter announced a crisis misinformation policy which would place a warning over (and not amplify or recommend) severely harmful crisis-related content that is deemed as misleading (as verified by “multiple credible, publicly available sources”) ([Roth, 2022](#)); whether such content will be identified by humans, algorithms, or a combination is not discussed. Such policies and algorithmic mechanisms define the media environments for content (and disinformation) creation and dissemination.

AI in content creation

Strides in machine learning contribute to the development of powerful artificial intelligence (AI) tools. Within machine learning, models have been developed to be trained on massive amounts of data through *unsupervised learning*, techniques through which the model detects underlying patterns *without* labeled data. Because unsupervised learning approaches do not require labeled data, these techniques are particularly powerful for dealing with big data ([Bommasani et al., 2021](#)). While the amount of training data and model effectiveness are not perfectly correlated, large amounts of data have contributed to powerful gains in more recent

models, as demonstrated by recent successes of large language models (Brown *et al.*, 2020). Once trained on data, AI models can then create novel output. AI models have gained increasing capabilities related to content creation, such as generating content through visual, verbal, and auditory modalities, as delineated in this review. This capacity for content creation can facilitate the generation of digital content at scale and the generation of content from reference images and text (as opposed to firsthand information). Thus, scholars must more closely examine how automated content generation and the creation of convincing, divisive content could affect online information ecosystems; especially if utilized by malicious actors, such technology can distort information transmission and perception and amplify the effects of disinformation.

Textual media. Natural Language Processing (NLP) is a rapidly evolving area of inquiry closely related to AI, driving advances related to models' abilities to complete human language tasks (Bommasani *et al.*, 2021). Recent strides within the field are related to large language models, which are trained on vast amounts of data to tune billions of parameters. Examples of these models include GPT-3 (Brown *et al.*, 2020), GPT-4 (OpenAI, 2023), and Gopher (Rae *et al.*, 2021). Such models have achieved promising results in a variety of language generation and classification tasks; research has demonstrated that these models are capable of generating text with humanlike quality in a variety of contexts. Studies have found: readers were unable to successfully differentiate between AI-generated and human-written content (Clark *et al.*, 2021), and individuals rated AI-generated public health messages to be of higher quality than Centers for Disease Control and Prevention (CDC) messages (Karinshak *et al.*, 2023). Such models have also been adopted in automated journalism, using algorithms to create news stories from data (Graefe, 2016). While current shortcomings include inconsistency in output quality (especially for applications requiring high-stakes information) (Karinshak *et al.*, 2023), and models' current abilities to handle variations in languages and language usage are less clear (Bommasani *et al.*, 2021), future automated textual content will become increasingly indistinguishable from human-authored text as these models grow in sophistication.

Visual media. AI models have also been applied to image-generation and modification tasks. Over the past decade, strides in generative networks (specifically generative adversarial networks (Goodfellow *et al.*, 2014)) have been applied for a variety of sophisticated image processing and generation tasks. The DRAW architecture (Gregor *et al.*, 2015) was used to generate images which were indistinguishable from the highly realistic originals in existing datasets. Another study generated images which respondents mistook as the real images 40% of the time (Denton *et al.*, 2015). Similarly, respondents were unable to effectively classify AI-generated images from the MNIST dataset (Salimans *et al.*, 2016). Models have also been developed to translate text to image, generating images of birds and flowers from descriptions (Reed *et al.*, 2016; Zhang *et al.*, 2017). Images with compelling detail have been generated more recently (Razavi *et al.*, 2019). This is illustrated by the emergence of deepfakes, defined as realistic, computer-generated images in a person's likeness (Kietzmann *et al.*, 2020). In addition to image generation, AI technologies can also be used to create and modify videos. For instance, researchers have used AI to generate videos from a provided caption (Pan *et al.*, 2017), and to project facial expressions onto a target image (Thies *et al.*, 2016).

Audio media. AI is also transforming audio tasks. It is being used for vocal pitch correction as an improvement to Auto-tune (Wager *et al.*, 2020). It also has been utilized for music creation, generating diverse and coherent songs in a variety of genres (Dhariwal *et al.*, 2020). Virtual assistants, such as Siri and Alexa, are designed to sound like natural language speakers. Audio deepfakes have also emerged, allowing the imitation, manipulation, or swapping of someone's voice (Kietzmann *et al.*, 2020).

AI in content dissemination

Practitioners already rely on modeling techniques to identify strategically meaningful audience segments from data, employing tools such as classic machine learning algorithms (e.g., k-means clustering), as well as more complex model architectures (e.g., deep neural networks). Such techniques are continuing to grow in sophistication, and existing research demonstrates that data mining and unsupervised learning techniques can be used to identify patterns, such as those spanning vast amounts of unstructured data related to consumer buyer behavior and preferences, not easily apparent to practitioners (Huang and Rust, 2021). By providing data related to social media usage, social media content, user profiles, and natural language on social media platforms to AI models, these models can determine meaningful audience segments.

Upon identification of segments and target users on digital platforms, automation techniques can then be employed to quickly disseminate content. Social bots, algorithms designed to emulate and interact with human users on social media platforms in a predefined manner (Ferrara *et al.*, 2016), are capable of altering online ecosystems through automatic content creation, as well as automated engagement with existing content (such as retweeting) (Ferrara *et al.*, 2016; Yuan *et al.*, 2019). Informed by segmentation insights and sophisticated content generation capabilities, such content can be extremely convincing, and concerning in malicious cases, such as the production or promotion of rumors, spam, malware, disinformation, slander, or noise (Ferrara *et al.*, 2016). Content is disseminated via the existing algorithmic landscape of dominant digital platforms, and affects the information users are exposed to. Shao *et al.* (2018) found that social bots play a disproportionate role in amplifying content which lacks credibility, creating early momentum prompting humans to engage and reshare as well. Regarding such human vulnerabilities, even when aware that algorithms are affecting the content displayed, this awareness usually does not evoke active adjustments in information-seeking behaviors (Swart, 2021); while the influence of any particular piece of content varies by a myriad of source-level (e.g., Hovland and Weiss, 1951; Sterrett *et al.*, 2019; Liu *et al.*, 2015), message-level (e.g., Mitchell, 1981; Greussing and Boomgaarden, 2019; Cauberghe *et al.*, 2009; Stone and Can, 2020), and recipient-level (Tynan and Drayton, 1987) factors, the content that users are exposed to and nature of processing (Petty and Cacioppo, 1986) have the potential to influence subsequent interpretations, attitudes, and behaviors.

AI-driven disinformation

AI technologies are achieving impressive generation capabilities, with AI and human-created content becoming increasingly difficult to distinguish (Verdoliva, 2020). Buchanan *et al.* (2021) evaluate the performance of large language models on a variety of disinformation-related tasks (e.g., narrative manipulation, narrative wedging), finding that while performance was imperfect, such models present real threats and powerful tools in the implementation of large-scale disinformation campaigns. In the short term, limitations of AI technologies, such as challenges with models generating text that is syntactically correct but semantically flawed (Sobieszek and Price, 2022), a lack of consistency in model performance when generating text for a specifically prompted purpose (Karinshak *et al.*, 2023), and cues such as digital imperfections or unnatural movements exposing deepfakes (Appel and Prietzel, 2022), limit the ability to seamlessly deploy these technologies in disinformation operations; however, AI technologies continue to rapidly improve. Given the already achieved levels of sophistication in content creation and rate of development (as delineated throughout this literature review), we draw upon these insights to propose that AI will approach perfect imitability at a content level—that images, videos, recordings, and other forms of media will eventually become perfectly indistinguishable by observation alone.

Together, AI creation and dissemination technologies can be leveraged to release and engage with malicious content at scale.

Content creation can include: creation of convincing headlines and articles, generation and manipulation of images and audio, making it challenging to distinguish between images and videos generated (Korshunov and Marcel, 2018), and manipulation of online conversations and ecosystems (e.g., automated engagement artificially inflating the popularity of divisive content) (Ferrara *et al.*, 2016; Yuan *et al.*, 2019). AI is able to generate convincing images of faces, enabling the creation of fake online accounts (Hill and White, 2020), which pose a threat to security, privacy, and journalism. Deepfakes, which can be used in the creation of malicious content using a person's likeness, present challenges related to disinformation and security, such as the creation of convincing videos of prominent figures and fake Internet accounts (Vaccari and Chadwick, 2020; Guardian News and Media, 2020). This production capacity can enable a greater barrage of convincing content on digital platforms and the creation of more targeted content (as variations of content can be produced almost instantly).

This automated, high-quality content creation can be paired with automatic dissemination through digital channels. Emerging AI technologies are capable of psychographic segmentation, leveraging machine learning models and drawing on extensive online data available on platforms such as Facebook (Ribeiro *et al.*, 2019). Just as targeted content can be generated, this content can also be automatically disseminated to specific audience segments. Such capacity for content generation introduces the ability to generate incredible amounts of copy almost instantly, facilitating the creation of disinformation at scale. Through microtargeting, malicious actors can employ such automation technologies to generate individualized content and target specific groups in a divisive manner (Kertysova, 2018); such content can then be amplified by automated social bots (Assenmacher *et al.*, 2020), which can be leveraged to artificially engage with divisive content, creating impressions of popularity and agreement and distorting perceptions of legitimacy (e.g., Shao *et al.*, 2018). Diaz Ruiz and Nilsson (2023) conceptualize this “echoing phase” in disinformation, emphasizing the role of supporting actors in propagating misleading narratives, contributing to their amplification. While much ambiguity surrounds the prevalence and measurement of social bots (Rauchfleisch and Kaiser, 2020), the capability for malicious actors to drive artificial engagement raises important considerations for future communication efforts.

This literature reflects the current state and trends in AI technologies, synthesizing how such developments relate to digital communication processes and disinformation. Together, such research demonstrates that AI can serve as an impactful tool augmenting and automating disinformation efforts. Next, we review the impact of these technologies on individual and organizational actors.

User perception and disinformation vulnerability

As AI-generated content is introduced and disseminated, understanding how psychological processes vary can enable practitioners to integrate organizational risk and crisis communication responses to address at-risk stakeholders' informational and behavioral needs, helping them cope with informational vulnerability and situational uncertainty. Information sharing decisions shape the reach of content and resulting information dissemination through social networks. Upon exposure to content, the receiver engages in conscious and subconscious information processing (Chaiken, 1980; Cappella *et al.*, 2014), which rapidly occurs as users digest information displayed in their feeds; the resulting interpretation affects what information is shared (Hunzaker, 2016) and how content spreads through users' social networks.

Factors affecting content perceptions include: source factors, message factors, and recipient factors. Perceptions related to source credibility, such as source trustworthiness, dynamism, and professionalism (Whitehead, 1968), affect subsequent interpretations (Hovland and Weiss, 1951; Sterrett *et al.*, 2019; Liu *et al.*, 2015). Online, users engage with accounts, such as celebrities, organizations, and influencers with large followings, personal connections (such as family and friends), and other users. Accounts with false or fake information can easily be created online, enabling source deception. At a message level, users are impacted by the topic (Mitchell, 1981), as well as the content's aesthetic (Greussing and Boomgaarden, 2019) and linguistic (Cauberghe *et al.*, 2009; Stone and Can, 2020) composition. Imperfections in content, such as misalignment between audio and visual or unnatural movements can signal manipulated content (Appel and Prielzel, 2022). Such factors affect the users' involvement and subsequent processing; higher involvement results in more active, deliberate processing, while lower involvement results in automatic processing (Petty and Cacioppo, 1986). Finally, recipients vary in their experiences, personalities, and beliefs. As reflected in marketing strategy literature, segmentation and targeting approaches can more specifically appeal to segments and contribute to their subsequent interpretations of information (Cahill, 1997); relevant variables for segmentation will vary by context, but can include "geographic, demographic, psychological, psycho-graphic or behavioural" dimensions (e.g., Tynan and Drayton, 1987).

Relating to perceptions of AI, existing research reveals that AI is capable of creating convincingly humanlike content in a variety of applications (e.g., Clark *et al.*, 2021). While AI-generated content and human-created content are generally comparable, nuances in quality perception vary. Human-created artwork achieved higher ratings for composition, degree of expression, and aesthetics (Hong and Curran, 2019); however, a digital piece developed by an artist with an AI tool (unknown by the judges) won the Colorado State Fair digital art contest (Kuta, 2022). AI-authored public health messages were also rated higher in message effectiveness and argument strength than CDC messages (Karinshak *et al.*, 2023) when AI was unknown as the author. However, perceptions tend to be less favorable when AI is a known actor. Individuals' interpretations were more negative when aware of AI assistance in interpersonal communication (Hohenstein *et al.*, 2021) and as the source of pro-vaccination public health messages (Karinshak *et al.*, 2023). Although AI is generally perceived more negatively when known, such literature demonstrates that in cases when people are unaware of its usage and ability to mimic human content, AI can produce effective content (both visual and textual). Thus, AI tools could be leveraged to generate content which is perceived as human.

Psychological processes affect users' vulnerability to disinformation. While source interpretations affect users' immediate responses to content, people tend to forget source while remembering the content itself over time (Hovland and Weiss, 1951); thus, users may remember deceitful content independent of its initial source. Furthermore, as an extension of the mere-exposure effect, research demonstrates that prior exposure can lead to increased perceptions of the accuracy of fake news (Pennycook *et al.*, 2018). Relating to exposure, malicious actors can increase the volume of content to create illusions of prevalence and concern (Schwarz and Jalbert, 2020). By distorting these perceptions, deceitful content may become more mentally available and salient to users (Schwarz and Jalbert, 2020), and it may create false impressions of public opinion and bandwagon pressures (Nadeau *et al.*, 1993). Such distortions can be convincing if the content does not inherently raise red flags. As posited by the truth-default theory, people perceive information with the default perspective that other users are communicating honestly, unless given a reason for suspicion (Levine, 2014). Together, this literature related to user perception provides insight into how stakeholders process the information disseminated via social platforms, and in particular, the vulnerabilities that emerge as AI-generated content is introduced; understanding how

platform characteristics and psychological processes interact is crucial for anticipating and mitigating the impact of AI-driven disinformation campaigns.

Organizational preparation and response: disinformation management

As practitioners seek to effectively communicate in technological environments which present novel challenges and evolving stakeholder engagement behaviors, strategies for disinformation management and response demand adaptation (e.g., [Hurley, 2023](#); [Guthrie and Rich, 2022](#)). Thus, we draw upon relevant public relations literature addressing organizational crises and response. Crisis misinformation is defined as “false information about any aspect of an ongoing crisis or any incorrect information that can lead to a crisis according to factual evidence from credible source(s) (e.g., the organization, news media, third-party experts, government agencies, and internal/external witnesses)” ([Jin et al., 2020](#)), which can lead to organizational crises ([Jin et al., 2020](#)). The SMCC model ([Liu et al., 2012, 2021](#)), which examines information flow and adoption in crises among interconnected media, publics, and organizations, can be applied to understanding misinformation and disinformation creation and dissemination. Both misinformation and disinformation result in the spread of false information, distinguished by intention (e.g., [van der Meer and Jin, 2022](#)). As the SMCC model has been applied in misinformation management, we extend these insights while assessing disinformation more specifically. The strategic considerations the SMCC model suggests ([Liu et al., 2012](#)) (i.e., crisis origin, crisis type, organizational infrastructure, message strategy, and message form) in managing a social-mediated crisis can be used by organizations under disinformation attack as a general guide for protecting business assets and stakeholders. Among these organizational considerations, having an organizational disinformation preparation and response infrastructure and optimizing disinformation countering/correcting message strategies are particularly essential. Theorizing and improving misinformation and disinformation management is at the forefront of the SMCC research advancement (see [Lu et al., 2022](#)).

Looking more closely at management and response strategies upon identification of troublesome content, organizational strategies (grounded in response strategies outlined in the SMCC model [see [Liu et al., 2023](#)] and specific strategies identified for managing crisis misinformation [see [Lu et al., 2022](#)]) which are intended to develop and foster disinformation resilience include: detection and monitoring, debunking efforts such as the release of corrective content, content flagging and removal, and prebunking strategies such as media literacy efforts.

Detection and monitoring. Existing strategies for disinformation detection include the identification of malicious actors and fact-checking. Machine-learning algorithms have been integrated into these processes, automating most content moderation with varying degrees of success ([Kertysova, 2018](#)). At a content level, detection methods began with human monitoring of content, but the exponential amount of online information makes manual fact-checking or review of all content slow, resource-intensive, and infeasible. Detection techniques have been developed for various forms of disinformation. For example, automatic facial detection techniques, as reviewed by [Gragnaniello et al. \(2022\)](#), have more recently been implemented to automatically differentiate real and synthetic images and videos (e.g., [Ciftci et al., 2020](#)). Similarly, scholars have recently assessed if NLP models could be effective detectors of automated content, finding that such models offer tremendous potential for future detection efforts ([Bommasani et al., 2021](#); [Wang et al., 2020](#)). However, a detector with universal applicability and reliable performance has yet to be developed, and detection becomes increasingly complex as AI models improve their generation capabilities ([Gragnaniello et al., 2022](#)). Researchers note the need for further sophistication of automated deepfake detection methods ([Korshunov and Marcel, 2018](#)). Currently,

combinations of automated and human review maximize monitoring effectiveness. For example, existing social media platforms allow users to report content that is spam, inappropriate, hateful, fraudulent, etc., calling attention to content and accounts that may be problematic. However, such an approach is reactive and likely to miss a great amount of troublesome content; for example, an estimated 90% of cyberbullying activity is unreported (Wang *et al.*, 2020). As researchers explore solutions which could stem from human-AI collaboration, another approach assesses how models can be leveraged to delegate decision making based on prediction certainty (Lai *et al.*, 2022); implementing such a technique would allow models to flag harmful content, handling straightforward cases while delegating less clear, more complex ones to humans.

Corrective communication. Jin *et al.* (2020) posited the concept of organizational corrective communication, which is defined as an organizational crisis response aimed to establish a prior message as misinformation (or disinformation if the intention is to manipulate or deceive). Organizational corrective communication was found to help debunk misinformation and correct misperceptions (Jin *et al.*, 2020; van der Meer and Jin, 2020). Scholars further argued that organizations should take more advantage in intercepting misinformation and disinformation spread by engaging social media users with corrective information (Jin *et al.*, 2020; Vijaykumar *et al.*, 2021). Current research on organizational corrective communication has focused on message characteristics, such as debunking strategies and sources (Jin *et al.*, 2020; van der Meer and Jin, 2020). On the one hand, compared to simple rebuttal, factual elaboration has been found to be a more effective correction message strategy in both health and organizational crisis settings. On the other hand, corrective information sources matter as well. In the context of organizational crisis, Jin *et al.* (2020) found that the availability of supportive supplemental information from a third actor (e.g., the voice and presence of a supportive employee on social media) can boost the organizational misinformation debunking efforts and correct misbeliefs about the organizational crisis.

Flagging and removal. Flagging serves as a means of allowing users to view content while providing some disclaimer. Much controversy surrounds ideal systems for flagging and removing content—visibly flagging content and content removal are heavily dependent upon authority and circumstance, and introduce a variety of social, ethical, and even political considerations. For example, Twitter announced a manipulated media policy in 2020 and delineated general criteria for the flagging and removal of content (Roth and Achuthan, 2020); most recently, Twitter announced a more specific plan for crisis misinformation removal (Roth, 2022). These announcements sparked online debate related to how this plan could be implemented in an ethical, unbiased manner (Karinshak, 2022) while preserving freedom of speech. Furthermore, the effectiveness of flagging is contested, as such flags draw additional attention to harmful content and research demonstrates that in certain cases, flagging can actually increase perceptions of legitimacy (Pennycook *et al.*, 2020). Future governmental and platform media policies will define the extent of flagging and removal. With the exception of the companies controlling media platforms, organizations do not possess direct authority to flag, moderate, or remove content. As these policies currently exist, removal has been limited. Thus, organizations should not expect the ability to flag or remove content in their responses.

Media literacy. While the strategies discussed thus far (detection and monitoring, corrective content, flagging and removal) are applicable at a content level, media literacy efforts represent a larger societal trend toward educated consumption of online media. In the context of the SMCC model, media literacy is one of several prebunking strategies that aim at building resilience before exposure to misinformation and disinformation attacks (Lu *et al.*, 2022). For example, Finland has recently integrated media literacy into their educational curriculum, emphasizing fact checking and increasing awareness of manipulative techniques online (such as clickbait, manipulated visual and audio, and fake news stories) (Mackintosh,

2019). Organizations can support media literacy efforts by supporting initiatives (e.g., through funding, and through their own content and engagement on social platforms). Such efforts can increase critical thinking in consumption and help mitigate the persuasiveness of disinformation. As [Hameleers \(2022\)](#) argued, news media literacy plays a key role in helping people separate truth from lies. News literacy messages are found to help enhance the effectiveness of misinformation correction by experts on Twitter ([Vraga et al., 2022](#)). However, an emerging concern is the impact of media literacy efforts in creating general skepticism and distrust in all media.

These practical and theoretical developments reflect another main approach identified by the SMCC scholars: motivating stakeholders to validate or vet the crisis information they are exposed to, which is the tenet of the stakeholder information vetting process ([Lu and Jin, 2020](#); [Lu et al., 2022](#)). Vetting, as a type of careful and critical examination for faults and errors ([Hawkins and Le Roux, 1986](#)), is especially essential for proactive disinformation attack prevention, preparation, and response. In the two-step information vetting model, [Lu and Jin \(2020\)](#) delineated: (1) in the primary vetting stage, stakeholders may access initial crisis information based on information consistency and feelings associated with information consumption and about themselves; (2) some stakeholders might be engaged in the secondary vetting stage, further assessing their feelings and might have second thoughts about their initial reactions and judgment. Such a vetting process presents a unique opportunity for organizations to facilitate and foster accesses for stakeholders to evaluate the facticity of received information (discerning truth from lies) and encourage them to counter-argue against misinformation and challenge disinformation, thus strengthening community information resilience.

Together, this literature review details the current state and trends in technological development, and the implications of these technologies for user engagement behaviors and organizational communication leveraging AI-mediated communication environments, especially in the effective management of AI-driven disinformation campaigns. The literature points to the growing sophistication of AI technologies, which can be applied for malicious purposes such as disinformation—understanding these technologies serves as a foundation for the development of forward-looking organizational response strategies in the evolving information environment. Informed by insights regarding the content amplification and stakeholder perceptual processes affected by these technologies and the strategies that have been adopted thus far, we apply these insights in conceptualizing future organizational response strategies.

A framework for organizational preparation for and response to AI-driven disinformation campaigns

Drawing upon this technological, perceptual, and organizational literature, we propose that factors differentiating AI-driven disinformation from previous forms of disinformation are *(1) the ability of malicious actors to disseminate and engage with a large volume of content through the coordination of automated systems and (2) an inability (of both humans and AI) to perfectly detect synthetic content and actors*. Looking ahead, grounded in the long-term assumption that applications of AI (e.g., deepfakes, verbal content, bot accounts) to produce disinformation will be able to perfectly imitate legitimate content, robust future organizational strategies will not be able to rely solely on detection strategies. These characteristics necessitate an updated approach for disinformation management. Given this characterization and the challenges associated with content detection and management, we instead assess the information *environment*, presenting a situational, actor-oriented perspective and discussing how to optimize reception of non-malicious content while disinformation is present.

With this literature-driven characterization of digital environments and integration with AI, we propose the following conceptual model for digital actors (see Figure 1), which adapts the SMCC framework to account for malicious and non-human actors. The model differentiates between the intentions of non-malicious and malicious actors and theorizes the interactions between actors. This model is devised from an organizational perspective with an emphasis on what to observe before and during engagement with various audiences (interacting with the organization, as indicated by the bidirectional arrow, and with each other within the “audience” arena, as indicated by the interconnected arrows with various directions). Observing the (dis)information transmission process systematically and making decisions on whether (and if so, how) to engage in informational intervention is essential for communication managers.

As illustrated, an organization releases content to their online audiences, and through an interactive process, their strategy for successive content is affected by audience response. Informed by the reviewed literature and building upon the SMCC, we identify key actors in digital environments as: creators (non-malicious and malicious) and receivers (human users and AI accounts). We define malicious actors as intentionally creating or propagating content designed to mislead, while non-malicious actors engage on platforms in the absence of malicious intentions; this differentiation is derived from the definition of disinformation (e.g., Fallis, 2009; Fallis, 2015; Ryan *et al.*, 2020; Guess and Lyons, 2020). Such roles are situational and dynamic (e.g., a human actor may be a non-malicious creator in one instance and simply a user [receiver] in another).

Creators, defined as any account that is actively releasing content relevant to the organization that shapes dialogue within the online ecosystem and the audiences’ resulting perceptions, are differentiated by *intention*, which can be malicious or non-malicious. Malicious and non-malicious actors include both human and AI entities; given anticipated similarities in content created by humans and generated by AI, these actors are not segmented by entity type, as literature demonstrates that human users cannot distinguish between human-written and AI-generated content in certain contexts (e.g., Clark *et al.*, 2021) and this framework proposes that these technologies are approaching perfect imitability.

Receivers, accounts that engage with content but do not create content, are differentiated by *entity type*—human users encompass all remaining human individuals, while AI accounts represent accounts that are capable of artificial engagement. The core difference among receivers is whether content has the potential to affect the entity’s subsequent attitudes and actions; this definition is based upon autonomy (or lack thereof). The model acknowledges

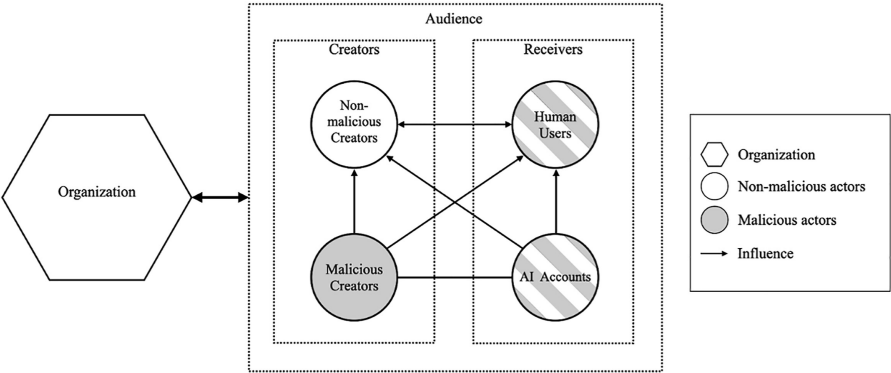


Figure 1.
A situational
perspective of actors
and their relative
influence in online
environments

Source(s): Created by authors

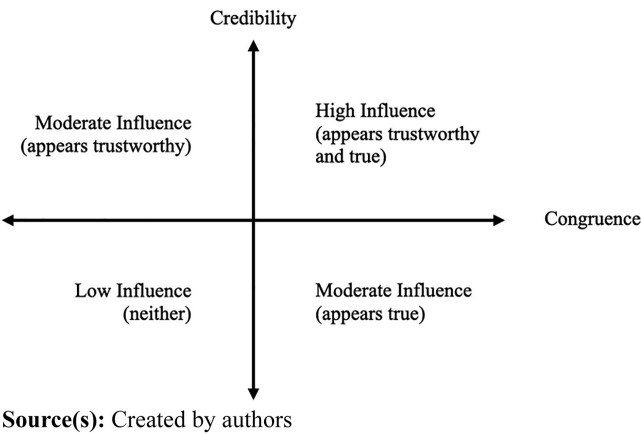
that both human and AI actors can be non-malicious or malicious. A small subset of human users may use their personal accounts for the purposes of propagating misleading information (e.g., in coordination with malicious human creators); thus, humans with malicious intentions can contribute to the dissemination of disinformation. Beyond content creation, AI accounts can also serve as a controllable means for malicious and non-malicious actors to engage in the environment. This definition is applicable so long as humans dictate the campaign objective, both when AI acts to create or augment content and dissemination, as the dissemination of (dis)information requires human direction and is applied to support human-driven agendas. The nature of AI account engagement is dependent upon the nature of the people controlling the engagement (which could be leveraged positively, negatively, or in a mixed fashion). In a disinformation context, these accounts can be leveraged to distort ecosystems (e.g., providing false legitimacy in the form of likes and followers). In contrast to human users, leveraging AI accounts to systematically propagate misleading information is significantly more scalable (and thus vulnerable to volume and engagement distortions, allowing a small number of malicious actors to exert vastly disproportionate influence when aided by AI tools) (e.g., [Shao et al., 2018](#)). Because these accounts appear on social media platforms as independent accounts, they are classified as a distinct set of actors.

The arrows between actors represent *relative influence*, defined as the impact of engagement behaviors (to include posting, sharing, commenting, liking, etc.) on actors' subsequent opinions and beliefs. Here we distinguish between influence and volume—while volume and influence are related, volume can be directly manipulated and distorted (as automated sources are capable of prolific content generation) while influence is a measure of impact (rather than quantity). *Relativity* refers to the comparison of influence between non-malicious and malicious content. The relationship between organizations and online audiences is reciprocal, as both parties perceive, interpret, and respond to feedback. Within digital audiences, malicious creators are characterized by outgoing links only; because their intent is malicious, the purpose of their engagement is to mislead (independent of the responses of other publics). Furthermore, AI accounts are also characterized by only outgoing influence, because as artificial entities, they do not hold attitudes and beliefs and are engaging as programmed (as opposed to interpreting content by other actors); their engagement affects how content gains momentum and is shared throughout networks. Because their actions are dictated by a human actor rather than the AI accounts' own attitudes and beliefs, although the accounts' actions may manifest differently depending upon other actors, its actions will support a consistent agenda (e.g., propagating a particular belief). In contrast, not adhering to a malicious agenda, non-malicious creators are gradually updating messaging strategies and non-malicious human users are refining their attitudes and beliefs as they receive and interpret content, indicated by incoming arrows (influence).

Furthermore, different pieces of content will be more or less influential, dependent upon source and message factors. Assuming that the information is of convincing quality (enabled by the technological advances reviewed above), we propose the following framework (see [Figure 2](#)) for determining content influence to differentiate among the perception and propagation of various types of disinformation.

Previous research has addressed the significance of perceptions of credibility and congruence in consumption (e.g., [Nadarevic et al., 2020](#)). Credibility refers to a users' impression of "believableness", based on source, message, and receiver factors (e.g., [Wathen and Burkell, 2002](#)). As human and computer communication become increasingly integrated, so do perceptions of credibility, impacted by factors including trust and algorithmic literacy ([Shin, 2021](#)). Congruence encompasses the contents' consistency with the attitudes and beliefs of a population (or subset of the population). Given these dimensions, the most influential disinformation falls into the top right quadrant, as this information appears trustworthy and true. Thus, this content is more likely to require an active response strategy, as it is more

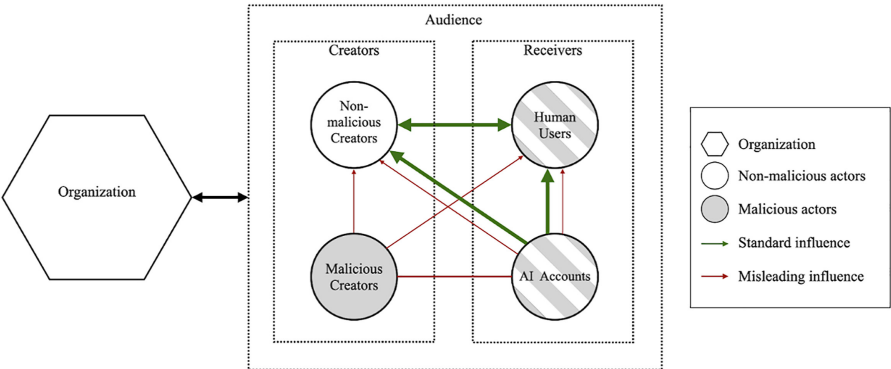
Figure 2.
Determining content
influence



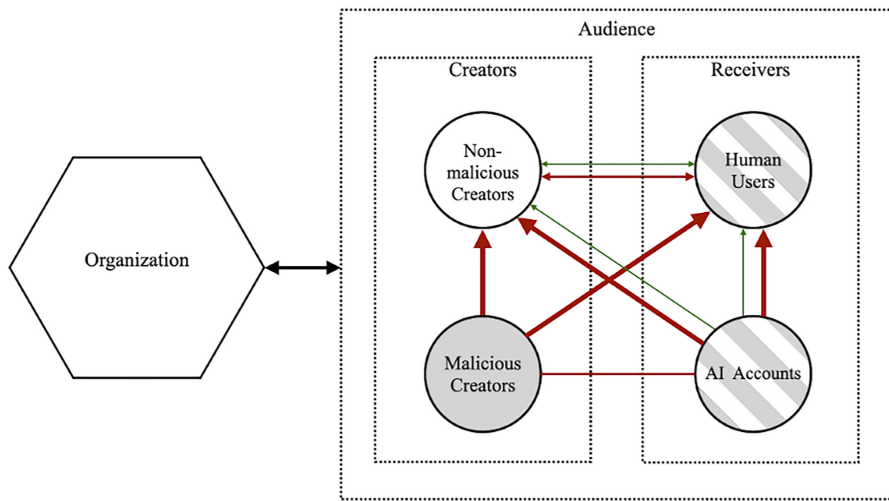
likely to gain traction and affect users' perceptions; practitioners should dedicate significant attention and strategic resources to handling such content. In contrast, in the bottom left, this low influence quadrant is low in credibility and congruence; thus, individuals are more likely to disregard the content, and even if initially influenced by the content, it is more easily debunkable. The remaining quadrants pose moderate influence and threat— while either low congruence or low credibility will partially discredit the content in the eyes of the receiver, the remaining, high dimension may compensate for the shortcoming, allowing the content to influence or partially influence receivers. Together, these quadrants of influence aid practitioners in prioritizing which content to address.

Given this conceptualization of actors and content influence, strategies for effective disinformation management will seek to maximize the relative influence of interactions between non-malicious creators and non-malicious human users. The following figures illustrate how relative influence contributes to the management versus spread of disinformation. In Scenario A (see [Figure 3](#)), the influence between non-malicious creators and non-malicious human users is dominant, minimizing the infiltration of disinformation. In contrast, when this relationship breaks down as demonstrated in Scenario B (see [Figure 4](#)), disinformation becomes dominant, gaining traction and affecting the strategies of non-

Figure 3.
Scenario A - dominant
non-malicious
influence



Source(s): Created by authors



Source(s): Created by authors

Figure 4.
Scenario B - dominant
malicious influence

malicious creators and attitudes and beliefs of non-malicious human users. Such infiltration, the sharing of divisive content among trusted, non-malicious actors, is the greatest threat and leads to harmful traction and sharing. This sharing process escalates as these non-malicious actors engage with and share the divisive content, contributing to virality.

Grounded in this conceptualization and call for maximizing influence among non-malicious actors in a manner that is robust against increased volume and highly realistic disinformation, we propose strategies for managing the relative influence of non-malicious content, as emergent from the reviewed literature. Related to social media platforms, platforms can affect display characteristics which are visible alongside content (such as flags, visibility of edits, account verifications, likes, etc.) and the recommendation algorithms feeding audience members content (e.g., recommended accounts to follow, and the extent to which the algorithm promotes variety versus similar content or contains content from accounts not directly followed). The remainder of this discussion centers on organizational strategies.

Related to organizational response, organizations can affect the following levers: proactively, their preexisting media presence and preparation, and reactively, response timing and messaging engagement strategy. Strategies for disinformation management have included: proactively developing a strong online presence (a clear mechanism for public communication) and following (Melchior and Oliveira, 2021), developing a strategic response plan, prebunking to provide users with a warning prior to exposure to disinformation in order to diminish its persuasiveness (Lewandowsky and Van Der Linden, 2021; Boman, 2021; Boman and Schneider, 2021), and quickly and publicly addressing disinformation with compelling corrective content (Jin *et al.*, 2020). The successful implementation of these strategies will become increasingly important as users face differentiating between legitimate and synthetic content and sources.

In addition to these classic disinformation management strategies, this AI-driven analysis brings to the focal point the importance of non-malicious creators supporting an organization's response (as opposed to the misleading content); organizations and malicious actors are competing for the support of influential non-malicious voices, who will largely affect how information is then perceived and disseminated. Thus, drawing upon

persuasion literature, this AI-driven model emphasizes the importance of fostering engagement that corroborates organizational content in the form of content from trusted, diverse sources across platforms and channels. Organizations can foster engagement (which creates algorithmic momentum) by strengthening their digital and non-digital PR efforts to relevant influencers, and by developing reactive content that is *shareable and prompts engagement* as well as effective (Shawky *et al.*, 2019). Strategies for doing so can include: (1) following and understanding trends on social media platforms; (2) differentiating and optimizing content for each social media platform (as the characteristics of impactful content differ on Facebook and TikTok); (3) creating content that is visually effective (with a strong sense of aesthetics) and digestible (with a clear purpose); and (4) fostering engagement (such as comments) through a question or call to action. Prebunking (e.g., Boman and Schneider, 2021) and debunking (Jin *et al.*, 2020) approaches can be integrated to enhance the effectiveness of the above recommended strategies and aid in strengthening these networks for information sharing.

Discussion

This framework illustrates the importance of influential creators in determining communication objectives. However, influencers are imperfect judges. Thus, it is essential for organizations to identify, establish, and strengthen relationships with important actors (including stakeholders), increasing perceptions of credibility and congruence, to ultimately increase the influence of the organizations' content. When perceptions of an organization's own credibility and/or congruence are low, its voice is weakened and it is more vulnerable to a disinformation crisis (as competing content can develop dominant influence instead, proposed by Scenario B). For example, controversy and skepticism surrounding genetically modified organisms (GMOs) has tremendous traction among the public; research has found that a small number of alternative health and pro-conspiracy sites ultimately received more total engagements on social media than sites generally considered relevant media outlets (Ryan *et al.*, 2020). In the case of Monsanto, one of the world's largest producers of genetically modified seeds, Monsanto CEO Hugh Grant joined news media discussion (e.g., "CBS This Morning" in December 2015) to address the ongoing debate over GMOs and respond to anti-GMO attacks by emphasizing Monsanto's commitment to innovation in agriculture and its attempt to enhance transparency in public communication. A 2016 Forbes article also discussed the complex and challenging nature of the informational attacks on GMO products and industry (Senapathy, 2016). However, it appears that most of the organizational responses to disinformation challenges rely primarily on mainstream media platforms with minimum interactions and engagement on social media where the disinformation is spread and gains increased traction. These observations and research findings exemplify the core proposal of the framework, that maximizing relationships and engagement are crucial when looking to the future of disinformation management.

Differentiating the impact of AI-driven disinformation from previously observed instances created by humans, while such instances low-tech disinformation can be harmful when widespread, the dangers and scale presented by AI-driven disinformation introduce greater potential for negative impact, as demonstrated by the existing literature and proposed framework. A limitation of this analysis is that examples of AI-driven disinformation campaigns in the corporate sector are currently limited. While a handful of tangential cases (e.g., deepfakes of public figures) provide insight to AI's capabilities, such as the realistic nature of deepfakes and content generation, the full force of such technologies has not yet been realized; future content could imitate company executives or employees, requiring preparedness from organizations and communication management professionals. Thus, while this research combines practical insights from communications and computer

science disciplines, it is anticipatory in nature. Future research can refine this framework and the development of response strategies, glean insights from specific instances of AI-driven disinformation, as researchers and practitioners observe and learn from the actual implementation of these technologies.

In characterizing emerging technologies, this research focused on how AI technologies may augment or distort information on major social media platforms as they currently exist. Another technology of growing importance is virtual and augmented reality, which introduces a myriad of new functionalities and has major implications for how future users will communicate and interact. Many practitioners and users question the need for such environments, and widespread appeal and adoption has not yet been achieved; regardless of what capacity these environments are utilized in the future, they will introduce novel mechanisms for information creation and dissemination, as well as ambiguity related to defining reality and truth. While the proposed model, extending SMCC literature and insights, represents communication processes alongside AI actors on social media platforms (limiting generalizability to future disruptive technologies), as new frontiers for digital interaction are introduced, future research can expand upon these insights and examine their resulting impact on communication and disinformation.

Furthermore, this research is focused on the corporate sector and response strategies within the control of an organization; however, social media platform design and algorithmic choices are also immensely important. Such design choices introduce tremendous ethical considerations (Voges and Peters, 2022), determining what content a user receives, options for engagement (e.g., liking, commenting), the content a user is subsequently exposed to, and whether content is flagged or removed. These platforms control the infrastructure for information creation and dissemination, and solutions for disinformation continue to be explored. Thus, we also call for continued research from a platform perspective regarding interface and algorithmic designs when detecting and responding to disinformation.

A broader, societal direction to consider is how societal norms online and offline affect disinformation consumption and sharing. For example, research finds that the social ramifications of sharing fake news (even when consistent with one's political ideology) serve as a tremendous deterrent in information sharing behaviors due to threat of reputational damage (Altay *et al.*, 2020). For users with accounts in their real name, accounts often serve as a tool for connecting professionally and/or socially with an extensive network (often, hundreds or thousands of connections); furthermore, for parasocial relationships and for relationships where connections no longer actively stay in touch, the account's content can serve as one of the main means for connections to form impressions. Thus, this social guardrail is beneficial in lowering the amount of disinformation shared. By normalizing and socially acknowledging media literacy behaviors such as fact checking, users can develop more robust routines for information consumption. Organizations, educators, and public figures can assist this normalization by publicly addressing such routines and remaining cognizant of how they communicate about disinformation from both human intelligence and AI, thus cocreating and improving a prebunking strategy based resilience system. Similarly, as technologies become more sophisticated, seeking to develop algorithmic literacy, "the combination of users' awareness, knowledge, imaginaries, and tactics around algorithms" (Swart, 2021, p. 2), can serve as a beneficial means for encouraging discretion in information consumption among publics. This can be accomplished through formal means (e.g., programs and educational curriculums), as well as informal means (e.g., TikTok videos).

Last but not least, by learning from the categorization of malicious and non-malicious actors in light of the role of AI, communication managers in different organizations can use the model this study posits to implement countermeasures for fighting AI-generated disinformation. In a recent PRNEWS article "How Corporate Communications Can Keep Up with Disinformation Attacks", Hurley (2023) suggested that the corporate communication

team might consider (1) partnering with a third-party organization that provides mitigation services, (2) closely monitoring the narrative and its reach across all platforms, (3) collaborating with influencers and stakeholders for accurate information dissemination (the importance of influencers in developing trust is also noted by [PRNEWS \(2022\)](#)), and (4) continuously building trust and credibility with customers and partners in order to strengthen the organization's overall digital resilience. The implementations of these practical recommendations, however, will need to be based on the identification of non-malicious versus malicious actors in order to choose which ones to collaborate or partner with and which ones to rebuke and fight back. Future research can provide additional practical insights for accurately recognizing actor intent. Our model provides a road map for communications managers to observe interactions among audiences and (dis)information flow in social and digital spaces they are in and choose strategic responses and countermeasures effectively and ethically.

Conclusion

The crux of this research clarifies how emerging technologies are affecting information (specifically disinformation) consumption and dissemination, and the implications for organizations and communication practitioners. As AI technologies become increasingly sophisticated, malicious actors can deploy such models for disinformation operations; thus, organizations must anticipate and be prepared to respond to disinformation in this evolving landscape. Guided by this proposed framework for influence (at actor and content levels), future communications efforts will necessitate the sharing and endorsement of information among trusted sources; additionally, consideration of disinformation credibility and congruence can aid practitioners seeking to prioritize resources in response to potentially harmful disinformation amongst vast amounts of content. These insights supplement existing research to aid researchers and practitioners in anticipating and combatting AI-driven disinformation.

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